Paper 74 — Shampoo

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Preconditioned stochastic tensor optimization

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The paper in one minute

Question: Can we precondition very large tensor-valued parameters without storing an enormous full gradient-covariance matrix?
Model/Method: Maintain a small preconditioner for each tensor mode, using accumulated gradient contractions rather than flattening everything.
Finding: Shampoo obtains regret guarantees in online stochastic convex optimization and shows competitive neural-network training performance in experiments.
Why we care: Gradient geometry can be exploited through tensor structure at far lower memory cost than an unrestricted full-matrix preconditioner.

1. Why coordinatewise adaptive scaling is not the whole story

SGD uses a scalar step $W^+ = W - \eta G$. AdaGrad-style diagonal scaling adjusts coordinates separately, but misses how gradient coordinates vary together. A full preconditioner might capture these correlations; for a matrix $W \in \mathbb{R}^{m\times n}$, however, flattening it creates a vector of length mn and a covariance-like matrix with $(mn)^2$ entries. Shampoo uses the fact that gradients *already have matrix or tensor shape*. It stores statistics along the left (row) and right (column) directions instead of forming the enormous flattened matrix.

2. Worked example: a 2×2 gradient

Let one gradient be

$$G = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Ignore the small identity damping term for this illustration and use one step of statistics:

$$L = GG^\top = \mathrm{diag}(4, 1), \qquad R = G^\top G = \mathrm{diag}(4, 1).$$

Shampoo's two-sided preconditioned update uses $L^{-1/4} G R^{-1/4}$. Since $4^{-1/4} = 1/\sqrt{2}$, we get

$$L^{-1/4} G R^{-1/4} = \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

The raw gradient's first entry was twice the second, but the structured preconditioner balances them in this example. It *does not* imply all matrix gradients become identity matrices.

The important memory saving
For a 1000×1000 weight matrix, a dense full flattened preconditioner would have 10^{12} entries. The two Shampoo factors together have about 2×10^6 entries, before implementation optimizations.

3. The matrix update, step by step

Algorithm 1 initializes $L_0 = \epsilon I_m$ and $R_0 = \epsilon I_n$. On each stochastic gradient G_t, accumulate

$$L_t = L_{t-1} + G_t G_t^\top, \qquad R_t = R_{t-1} + G_t^\top G_t,$$

then update

$$\boxed{W_{t+1} = W_t - \eta L_t^{-1/4} G_t R_t^{-1/4}.}$$

Why the negative quarter powers? Both left and right factors contribute to the overall scaling. When interpreted through vectorization and a Kronecker-product preconditioner, the combination behaves as a structured analogue of the inverse square-root scaling in matrix AdaGrad. These fractional powers are spectral matrix powers, not elementwise powers.

4. Extension from matrices to tensors

For a k-way tensor with mode lengths $n_1, \ldots, n_k$, Shampoo maintains one $n_i \times n_i$ preconditioner per mode. Each factor is formed by contracting gradient tensors over all the other modes. This brings memory closer to $\sum_i n_i^2$ instead of $(\prod_i n_i)^2$. The algorithm's tensor-mode contractions and fractional powers are designed to respect the parameter's structure.

What makes the proof work?

Theoretical results use online mirror descent and regret bounds in a *convex* setting. The paper's matrix analysis uses a Kronecker-product argument and matrix inequalities (especially Lemmas 8–9) to relate the structured factors to a more expensive full-matrix update. The structure is not arbitrary: the relevant preconditioners must be positive definite (the ϵI initialization helps), and tensor ranks enter the regret analysis.

Different use of the word "second order"
Shampoo uses accumulated outer products of *gradients*, not the exact loss Hessian. It is a structured adaptive preconditioner, not a Newton step or a guaranteed negative-curvature detector.

5. Results, contributions, and limitations

Contributions. (i) Tensor-mode preconditioning with manageable state; (ii) an elegant two-sided matrix update; (iii) convex online regret guarantees grounded in matrix trace inequalities; and (iv) experiments on modern-for-2018 neural-network training tasks. **Empirical result.** The paper reports faster optimization than standard first-order baselines in selected neural-network workloads, with practical per-step costs that can be comparable under its implementations. Those empirical observations are separate from the convex regret guarantees.

6. Important assumptions and caveats

- The formal guarantee is online/stochastic *convex* optimization; it does not prove global convergence for arbitrary deep neural networks.
- A dense eigendecomposition or matrix-root computation for each mode can be expensive; actual efficiency depends on update frequency, block sizes, and system implementation.
- Tensor modes encode useful structure, but a Kronecker approximation cannot capture every correlation in a flattened gradient.
- Neither strict-saddle escape nor improved test accuracy is guaranteed by the foundational theorem.

7. Quick reference

Matrix parameter	$W \in \mathbb{R}^{m\times n}$; gradient G has same shape.
Left statistic	$L_t = \epsilon I + \sum_{s\le t} G_s G_s^\top$.
Right statistic	$R_t = \epsilon I + \sum_{s\le t} G_s^\top G_s$.
Update	$W^+ = W - \eta L^{-1/4} G R^{-1/4}$.
Memory	Two factors: $O(m^2 + n^2)$ versus $O(m^2 n^2)$.
Theory scope	Convex online regret; deep-network speedups are experimental.

Survey bridge / Papers Master note

Shampoo uses row/column or tensor-mode gradient statistics to build memory-efficient structured preconditioners. The original theory establishes convex online regret guarantees, while neural-network gains are empirical. Cite in the adaptive/preconditioned dynamics discussion, not as a general saddle-escape theorem.